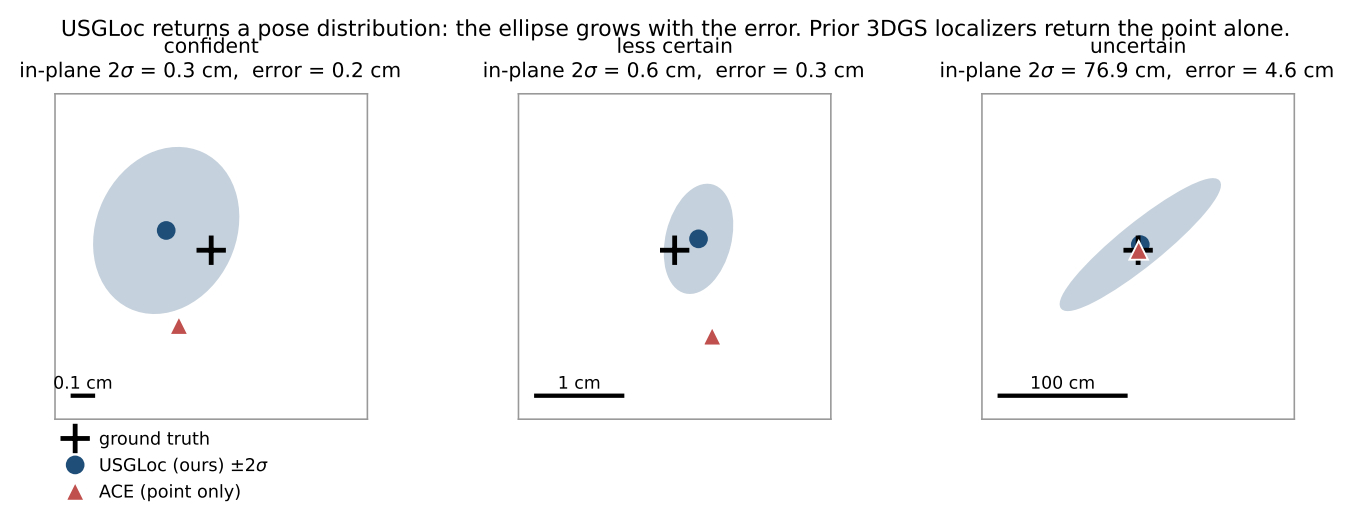
USGLoc returns a pose distribution: the ellipse grows with the error. Prior 3DGS localizers return the point alone.
confident
in-plane 2σ = 0.3 cm, error = 0.2 cm
less certain
in-plane 2σ = 0.6 cm, error = 0.3 cm
uncertain
in-plane 2σ = 76.9 cm, error = 4.6 cm
0.1 cm
1 cm
100 cm
ground truth
USGLoc (ours) ±2σ
ACE (point only)